

CO3 AT3 Tree Traversal

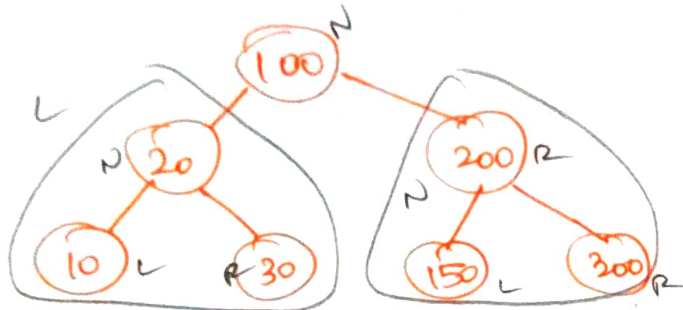
Course Code: CFA0311

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Course Name: Data Structure.

Reg No: 192570082

1Q.

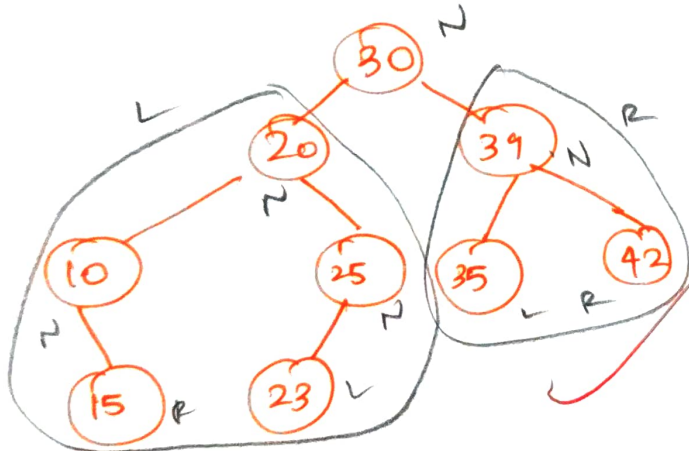


Inorder: LNR $\rightarrow$ $\frac{10 \ 20 \ 30}{L} \quad \frac{100}{N} \quad \frac{150 \ 200 \ 300}{R}$

Preorder: NLR $\rightarrow$ $\frac{100}{N}, \frac{20, 10, 30}{L}, \frac{200}{R} \quad \frac{150 \ 300}{R}$

Postorder: LRN $\rightarrow$ $\frac{10, 30, 20}{L} \quad \frac{150, 300}{R} \quad \frac{200}{R} \quad \frac{100}{N}$

2Q.

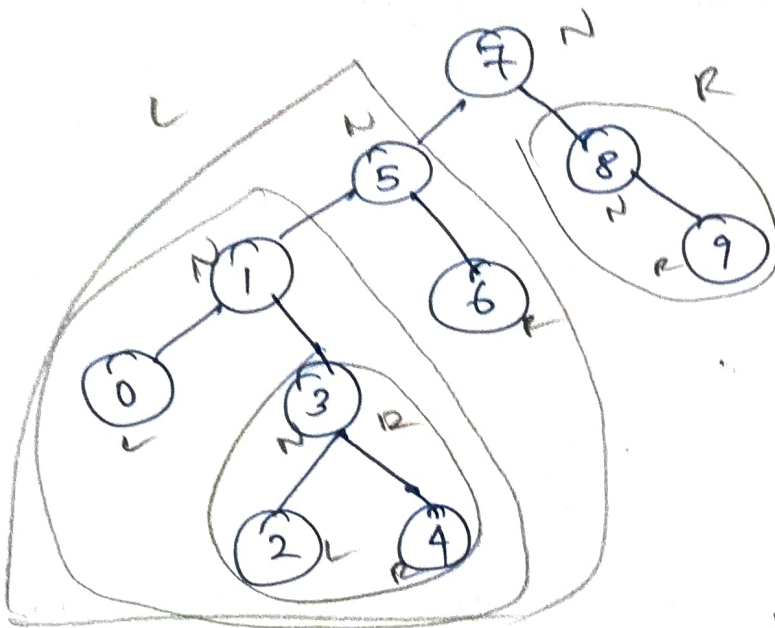


Inorder: LNR $\rightarrow$ $\frac{10 \ 15 \ 20 \ 23 \ 25 \ 30}{L} \quad \frac{35 \ 39 \ 42}{R}$

Preorder: NLR $\rightarrow$ $\frac{30}{N}, \frac{20, 10, 15, 25, 23}{L}, \frac{39, 35, 42}{R}$

Postorder: LRN $\rightarrow$ $\frac{15, 10, 23, 25, 20}{L} \quad \frac{35, 42, 39}{R} \quad \frac{30}{N}$

Q3. 7 5 18 3 6 0 9 4 2 → Binary search Tree.

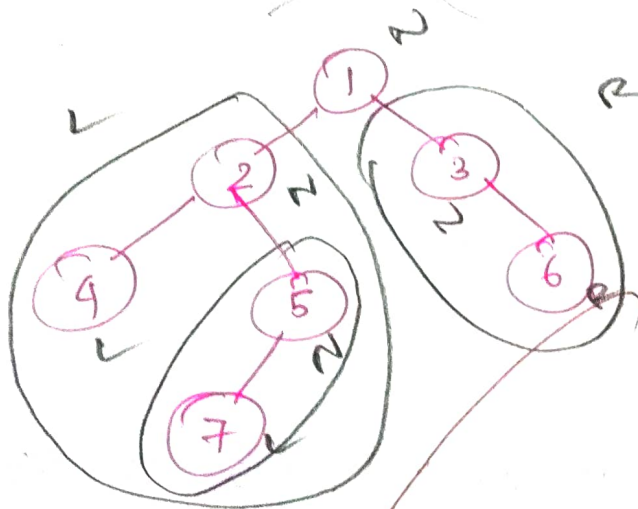


Inorder: LNR → 0 1 2 3 4 5 6 7 8 9

Preorder: NLR → 7 5 1 0 3 2 4 6 8 9

Postorder: LRN → 0 2 4 3 1 6 5 9 8 7

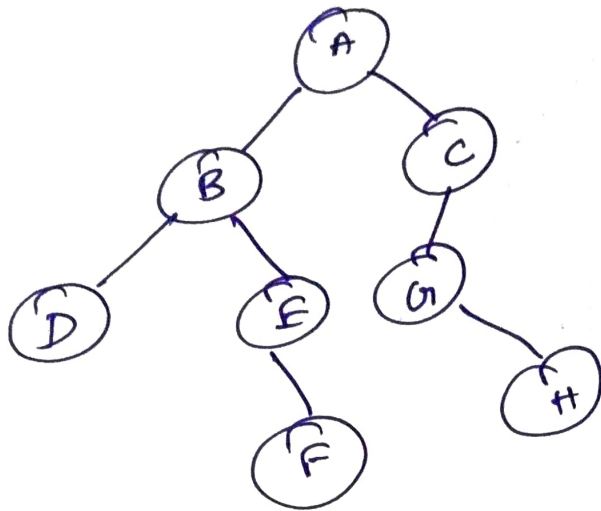
Q5.



Inorder: LNR → 7 4 2 5 1 3 6

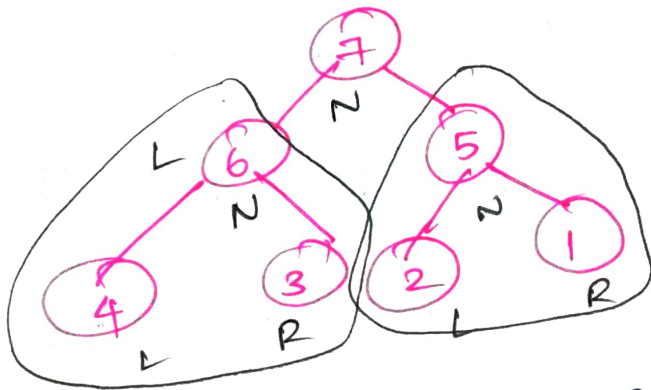
Preorder: NLR → 1 2 4 7 5 3 6

Postorder: LRN → 7 4 5 2 6 3 1



Preorder: LRN $\rightarrow$ $\frac{D \ F \ E \ B}{L} \ \frac{G \ H \ C}{R} \ \frac{A}{N}$

9Q

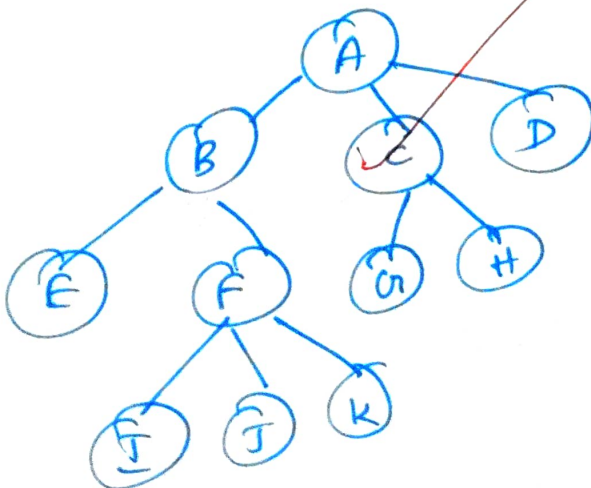


Inorder: LNR $\rightarrow$ $\frac{4 \ 6 \ 3 \ 7}{L \ N} \ \frac{2 \ 5 \ 1}{R}$

Preorder: NLR $\rightarrow$ $\frac{7}{N} \ \frac{6 \ 4 \ 3}{L} \ \frac{5 \ 2 \ 1}{R}$

Postorder: LRN $\rightarrow$ $\frac{4 \ 3 \ 6}{L} \ \frac{2 \ 1 \ 5}{R} \ \frac{7}{N}$

10Q



Inorder : LNR $\rightarrow$ Not defined

Preorder : NLR $\rightarrow$ A B E F I J K C G H D

Postorder : LRN $\rightarrow$ E I J K F B G H C D A.

